

Complete Classification for a Power–Congruence Norm Identity

Discovery packet for arXiv:2403.07341

13 August 2026

Status. Candidate full theorem, likely valid, pending human review. The problem is present in the raw arXiv TeX only *after* `\end{document}` and is absent from the rendered paper. The theorem below therefore solves an orphaned drafting remnant, not a question visible in the published PDF.

1 Source problem and provenance

Hatori and Oi’s *Non-linear characterization of Jordan $\ast$ -isomorphisms via maps on positive cones of $C^\ast$ -algebras*, arXiv:2403.07341, studies positive cones of unital $C^\ast$ -algebras [1]. Its raw source contains the following final problem. For $a \in \mathcal{A}_+^{-1}$ and $p > 0$, put

$$\phi_{a,p}(x) = (a^{p/2} x^p a^{p/2})^{1/p}, \quad x \in \mathcal{A}_+^{-1}.$$

Determine when

$$\|\phi_{a,p}(x)a^{-1}\| = \|x\| \quad (x \in \mathcal{A}_+^{-1}). \quad (1)$$

The raw note gives two endpoint answers: every a works for $p = 2$, whereas for $p = 1$ the condition is equivalent to centrality of a .

The source’s `\end{document}` occurs immediately before this “Final remark.” Consequently, the 25-page arXiv PDF ends with the bibliography and does not display the problem. The local file `source_problem_excerpt.txt` preserves the exact raw-source passage. The file `figures/raw_source_problem.png` is a screenshot of that block; there can be no honest crop of the problem from `source_paper.pdf`.

2 Proof intuition

The norm identity looks nonlinear, but positive elements turn a norm bound into an order bound. After squaring and changing variables, it says exactly that raising to a power $q = 2/p$ preserves *and reflects* every comparison with the ray through $c = a^p$:

$$z \leq tc \iff z^q \leq t^q c^q.$$

If $q \neq 1$, one direction of this equivalence gives local monotonicity of a strictly convex power function at a suitable inverse. Nagy’s centrality theorem then forces c , and hence a , to be central. The exceptional value $q = 1$ is precisely $p = 2$, where the identity follows by taking the square of the norm. This mechanism also explains why the two cases already recorded in the raw source are the two branches of the complete answer.

3 Formal result

Throughout, $\mathcal{A}$ is a unital C^* -algebra, $Z(\mathcal{A})$ is its center, and $\mathcal{A}_+^{-1}$ is its cone of positive invertible elements.

Lemma 1 (lower local power rigidity). *Let $d \in \mathcal{A}_+^{-1}$ and $r > 1$. If*

$$0 < w \leq d \implies w^r \leq d^r \quad (w \in \mathcal{A}_+^{-1}),$$

then d is central.

Proof. Put $b = d^{-1}$. If $v \in \mathcal{A}_+^{-1}$ and $b \leq v$, then inversion reverses the order, so $v^{-1} \leq d$. The hypothesis gives $v^{-r} \leq d^r$; another inversion gives

$$b^r = d^{-r} \leq v^r.$$

Thus the strictly convex increasing function $f(t) = t^r$ on $(0, \infty)$ is locally monotone at b : $b \leq v$ implies $f(b) \leq f(v)$. By Nagy's centrality theorem [2, Theorem 1], b is central. Hence so is $d = b^{-1}$. $\square$

Theorem 1 (complete classification). *Let $a \in \mathcal{A}_+^{-1}$ and $p > 0$. Then (1) holds for every $x \in \mathcal{A}_+^{-1}$ if and only if*

$$p = 2 \quad \text{or} \quad a \in Z(\mathcal{A}).$$

Proof. First assume (1). Set

$$c = a^p, \quad q = \frac{2}{p}.$$

For arbitrary $y \in \mathcal{A}_+^{-1}$, substitute $x = y^{1/p}$ and square the norm in (1). Since $c^{1/2} = a^{p/2}$ and $c^{-q/2} = a^{-1}$, this gives

$$\left\| c^{-q/2} (c^{1/2} y c^{1/2})^q c^{-q/2} \right\| = \|y\|^q. \quad (2)$$

We claim that for every $z \in \mathcal{A}_+^{-1}$ and $t > 0$,

$$z \leq tc \iff z^q \leq t^q c^q. \quad (3)$$

Indeed, write $z = c^{1/2} y c^{1/2}$. All elements whose norms occur below are positive, so (2) yields the chain of equivalences

$$\begin{aligned} z \leq tc &\iff y \leq t\mathbf{1} \iff \|y\| \leq t \\ &\iff \left\| c^{-q/2} z^q c^{-q/2} \right\| \leq t^q \iff c^{-q/2} z^q c^{-q/2} \leq t^q \mathbf{1} \\ &\iff z^q \leq t^q c^q. \end{aligned}$$

Suppose first that $q > 1$. Taking $t = 1$ in (3) shows that $z \leq c$ implies $z^q \leq c^q$. Lemma 1, with $d = c$ and $r = q$, makes c central.

Now suppose that $0 < q < 1$. Put $r = 1/q > 1$ and $d = c^q$. If $0 < w \leq d$, set $z = w^r$. Then $z^q = w \leq c^q$. The reverse implication in (3) gives $z \leq c$; equivalently,

$$w^r \leq c = d^r.$$

Lemma 1 makes d central, and consequently $c = d^r$ is central. In both nonexceptional cases $q \neq 1$, functional calculus gives $a = c^{1/p} \in Z(\mathcal{A})$. Since $q = 1$ exactly when $p = 2$, necessity is proved.

For sufficiency, first let a be central. It commutes with every positive x , so

$$\phi_{a,p}(x) = (a^p x^p)^{1/p} = ax, \quad \phi_{a,p}(x)a^{-1} = x,$$

for every $p > 0$. Finally, if $p = 2$, no commutativity is needed: writing $B = (ax^2a)^{1/2}$,

$$\|Ba^{-1}\|^2 = \|(Ba^{-1})^*(Ba^{-1})\| = \|a^{-1}B^2a^{-1}\| = \|x^2\| = \|x\|^2.$$

This is (1) and completes the proof. □

4 Verification and adversarial checks

- The proof uses both directions of (3). The reverse direction is essential when $p > 2$ (equivalently $q < 1$).
- All substitutions are bijections of $\mathcal{A}_+^{-1}$: $x \mapsto x^p$ and $y \mapsto c^{1/2}yc^{1/2}$. Thus no positive invertible test elements are lost.
- Lemma 1 invokes Nagy's theorem only for the strictly convex power t^r , $r > 1$, on the open interval $(0, \infty)$, exactly within its hypotheses.
- Random 2×2 tests with noncentral diagonal a and noncommuting x fail away from $p = 2$ and agree at $p = 2$. The script `verify_examples.py` is a non-proof sanity check.

5 Novelty, limitations, and review recommendation

A bounded search on 13 August 2026 covered the local run indexes and parsed arXiv corpus; exact formulas and phrases from the raw problem; the paper title and authors; and close web searches for the power-congruence identity. No later paper or explicit resolution of this exact classification was found. Nagy's 2019 theorem is a decisive prior ingredient, but it does not state the norm classification; the reduction (3) is the new synthesis. This is not an exhaustive novelty claim.

The principal provenance limitation is unusually important: the problem is not in the rendered source paper. A reviewer should (i) check the norm-to-order equivalence, especially the $q < 1$ branch; (ii) confirm the use of Nagy's Theorem 1; and (iii) decide whether an unrendered arXiv drafting remnant merits presentation as an open-problem solution. Human review is not yet complete.

References

- [1] O. Hatori and S. Oi, *Non-linear characterization of Jordan $*$ -isomorphisms via maps on positive cones of C^* -algebras*, arXiv:2403.07341 (2024).
- [2] G. Nagy, *Characterizations of centrality in C^* -algebras via local monotonicity and local additivity of functions*, Integral Equations Operator Theory **91** (2019), Paper No. 28, doi:10.1007/s00020-019-2526-2.